Multi-Party Authorization Proxy

Georgia Tech — CS 6727 Cybersecurity Practicum

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AI Use: Project, deck, and demo created with help from Claude & GPT 5.6; research, design, and all project decisions are my own.

Multi-party authorization is an underused security control

WHAT IS IT?

Requiring more than one person to approve before an action goes through.

WHERE YOU ALREADY SEE IT

Crypto multisig

moving funds

AWS

backup vaults · key imports

Google Cloud

privileged IAM grants

Azure

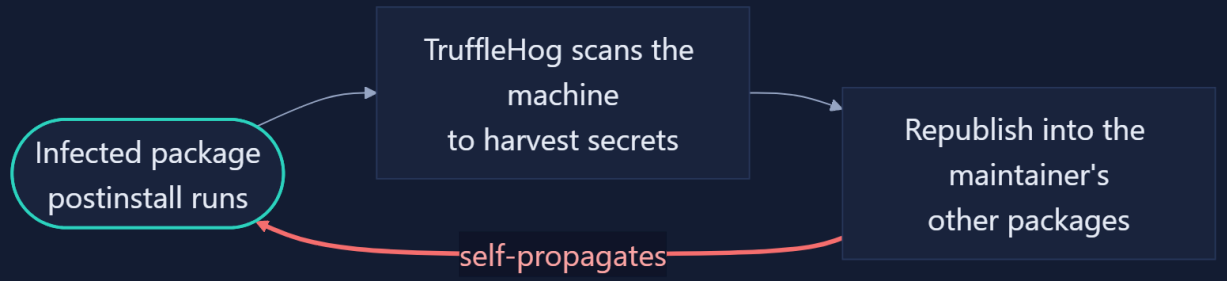
backups · device configs

My project: a general-use multi-party authorization proxy

My headline use case: package publishing

Case Study

September 14, 2025 - An npm supply-chain worm begins.



1

stolen publish credential — all it needed to start

~500

packages compromised in roughly a day

2M+

weekly downloads on a single infected package

They named it Shai-Hulud

The worm dumps every stolen secret into a public GitHub repo it names *Shai-Hulud*, after the sandworm from *Dune*.

Dune: Part Two © Warner Bros. Pictures

WHAT NPM & GITHUB BUILT IN RESPONSE

- Mandatory 2FA on local publishing
- Granular tokens, 7-day lifespans
- Trusted Publishing, no long-lived tokens
- FIDO hardware keys replacing TOTP

Every fix hardened who publishes — and it kept surfacing.

Sep 2025
Patient Zero · ~500 packages

Nov 2025
"Second Coming" · ~700 packages · 25k+ repos

Spring 2026
TanStack · valid SLSA L3

Jun 2026
Red Hat packages

Jul 2026
AsyncAPI · bypassed all review

Not just npm. Not just one worm

Every control the industry built to stop a poisoned release, against every way a release gets poisoned.

Control and the decision it actually gates	Stolen credential Shai-Hulud	Trusted insider XZ backdoor	Compromised CI poisoned pipeline	Direct publish skip repo + CI
Mandatory 2FA / MFA proves login identity	~	✗	✗	✗
Trusted Publishing (OIDC) proves publish identity	✓	✗	✗	~
Build provenance (SLSA) attests build integrity	✗	✗	✗	✗
Required reviews + branch protection gates the repo merge	✓	~	✗	✗
CI/CD deployment gates gates the pipeline deploy	✓	~	~	✗
Artifact-repo promotion gates internal promotion	✓	~	~	✗

✓ covers it · ~ partial · ✗ misses it

The missing layer: authorization

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Required reviews + branch protection gates the repo merge	✓	~	✗	✗
CI/CD deployment gates gates the pipeline deploy	✓	~	~	✗
Artifact-repo promotion gates internal promotion	✓	~	~	✗
The proxy authorizes the release itself	✓	✓	✓	✓*

✓ covers it · ~ partial · ✗ misses it · ✓* under one deployment precondition, revisited later

How the proxy authorizes a release

It sits in front of the registry: an upload is *held, not published*, until the package's owners authorize the exact artifact.

1

Request

Published via **twine** — to the proxy, not the registry.

2

Hash-bind

Artifact **hashed** and held; the approval is **bound** to it.

3

Approve

Each owner **re-auths** and **signs** their vote.

4

Quorum

Waits for **m-of-n**. One denial stops it.

5

Publish

Hash **re-checked**, then released to PyPI.

> The owners approve **one exact artifact** — and that's the artifact that ships.

Demo

Aren't we adding new risks?

Yes. One gate in front of every release concentrates risk — the biggest new threat the project has. So here's the *whole* surface it introduces: all 33 threats, catalogued.

CONCENTRATION OF RISK

CODE-1 · CODE-2 · CRYPTO-1 · HOST-1 · HOST-2 · HOST-3 · HOST-4 · IDENT-1 · IDENT-6

HUMAN FACTOR

VOTE-1 · VOTE-2 · VOTE-3 · VOTE-4 · VOTE-5 · IDENT-2 · IDENT-4

AVAILABILITY

DOS-1 · DOS-2 · DOS-3 · DOS-4 · IDENT-5

ALSO INTRODUCED

INFO-1 · PUB-1 · PUB-2

IMPROVED

CORE-1 · CORE-2 · CORE-3 · CORE-4 · HOST-5

INHERITED

CRYPTO-2 · CRYPTO-3 · IDENT-3 · PUB-3

Concentration — the gate is one target · Human factor — fatigue · coercion · replay · Availability — the gate as a jam point

Other — outside the three themes · Improved — threats the proxy makes better · Inherited — out of scope, the registry's surface

It only works if it's deployed right

The proxy's guarantees rest on a few steps it *can't enforce for you*. Two carry the most weight.

Revoke every pre-existing publish token. The whole guarantee rests on the proxy holding the *only* credential that can upload — leave an old one live and an attacker skips the gate entirely.

PUB-2 · proxy bypass

Choose the quorum and the co-owners deliberately. High enough to mean something, low enough that losing one approver can't freeze a release — and choose co-owners that are unlikely to collude.

DOS-3 · DOS-4 · CORE-3

… ~45 more — the full operator checklist ships with the proxy

What I'm not claiming

A fully colluding quorum

If every required co-owner conspires, the proxy approves the release — it did exactly what they told it to. Raising the bar from one person to several was never a promise to survive everyone you trusted turning at once.

CORE-3 · out of scope by design

The registry's own surface

The proxy sits *in front of* a registry it doesn't own, so it inherits that registry's weaknesses. Take over the PyPI account through its recovery flow and you never touched the proxy at all.

PUB-3 · inherited

Future Work

Where this use case goes next

Build it into the registry. As a native feature, the biggest risk the proxy introduced dissolves — the gate *becomes* the registry, already the juiciest target in the ecosystem. Trusted Publishing is proof the community will change the publish path when the case is strong.

Multi-party authorization as a whole

An option in more places. It belongs anywhere one stolen credential can trigger something you can't take back: a production deploy, a root cloud account, a wire transfer.

For anything you can't take back, one stolen credential shouldn't be enough — and with Multi-Party Authorization, it doesn't have to be.

LET'S CONNECT - IAN BARISH